

Skills Summary

Volume, Conversions & Angles

Use this reference while completing your worksheet or when studying.

Skill 1 — Volume of a Box

Formula: $V = \ell \times w \times h$ (cubic units, e.g. cm^3).

Think in layers: the base holds $\ell \times w$ unit cubes, and there are h layers. Missing a dimension? Divide the volume by what you know.

Example: Find the volume of a $6 \times 3 \times 2$ cm box.

Step 1. A box's volume counts unit cubes, so multiply length, width, and height, and answer in cubic centimeters.

Step 2. Base layer: $6 \times 3 = 18$ cubes; 2 layers make 36.

$$\Rightarrow V = 36 \text{ cm}^3$$

Try it: Find the volume of a $4 \times 4 \times 2$ cm box.

check: 32 cm^3

Skill 2 — Converting Units

Key facts: m $\rightarrow$ cm: $\times 100$ • km $\rightarrow$ m: $\times 1000$ • kg $\rightarrow$ g: $\times 1000$ • L $\rightarrow$ mL: $\times 1000$ • hours $\rightarrow$ minutes: $\times 60$.

Going from a *bigger* unit to a *smaller* one multiplies; going back divides.

Example: Convert 3.5 m to centimeters.

Step 1. Meters are bigger than centimeters, so the count of units grows — multiply by the number of centimeters in one meter.

Step 2. $3.5 \times 100 = 350$.

$$\Rightarrow 3.5 \text{ m} = 350 \text{ cm}$$

Try it: Convert 4 kg to grams.

check: 4000 g

Skill 3 — Finding a Missing Angle

Key facts: a right angle is 90° • a straight line is 180° • a full turn is 360° .

Angles *add*: when a big angle is split, the parts sum to the whole, so the missing part is whole – known part.

Example: Two angles make a straight line; one is 140° . Find the other.

Step 1. The two parts must add to the straight line's whole, so subtract the known part from that whole.

Step 2. $180 - 140 = 40$.

$$\Rightarrow 40^\circ$$

Try it: A right angle is split; one part is 25° . Find the other part.

check: 65°

(!) **Watch out:** before computing, make every measurement use the *same unit* — you cannot subtract meters from kilometers. Convert first, then compute.